TEST-HQ4KO1

FUNCTION / METHOD NAME: CALCULATETOTAL() TEST SUITE

Fri, 13 Mar 2026 07:53:00 GMT

Function / Method Name: calculateTotal().test.py

PYTHON | PYTEST | INTERMEDIATE | TARGET: 86%

```
"c-kw">def calculate_total(price: float, tax: float) → float:
    "c-kw">return price + (price * tax)
```

HAPPY PATH TESTS

3 tests

```
"c-kw">class TestFunction / Method Name: calculateTotal():
    "c-kw">def test_returns_expected_result_for_valid_input(self):
        result = Function / Method Name: calculateTotal()('valid_input')
        "c-kw">assert result is not None

    "c-kw">def test_handles_typical_use_case(self):
        result = Function / Method Name: calculateTotal()({'id': 1, 'name': 'test'})
        "c-kw">assert result['success'] is True

    "c-kw">def test_returns_consistent_results(self):
        "c-kw">assert Function / Method Name: calculateTotal()('input') == Function / Method Name:
calculateTotal()('input')
```

EDGE CASE TESTS

3 tests

```
"c-kw">class TestFunction / Method Name: calculateTotal()EdgeCases:
    "c-kw">def test_handles_empty_string(self):
        result = Function / Method Name: calculateTotal()('')
        "c-kw">assert result is not None

    "c-kw">def test_handles_zero_value(self):
        result = Function / Method Name: calculateTotal()(0)
        "c-kw">assert result is not None

    "c-kw">def test_handles_large_input(self):
        large = 'a' * 10000
        result = Function / Method Name: calculateTotal()(large)
        "c-kw">assert result is not None
```

NULL / UNDEFINED TESTS

3 tests

```
"c-kw">import "c-kw">pytest

"c-kw">class TestFunction / Method Name: calculateTotal()NullCases:
    "c-kw">def test_handles_none_input(self):
        result = Function / Method Name: calculateTotal()(None)
        "c-kw">assert result is None or result is not None  "c-kw">class="c-cmt"># graceful

    "c-kw">def test_raises_on_missing_required_param(self):
        "c-kw">with "c-kw">pytest."c-kw">raises(TypeError):
            Function / Method Name: calculateTotal()()
```

ERROR / EXCEPTION TESTS

3 tests

```
"c-kw">import "c-kw">pytest

"c-kw">class TestFunction / Method Name: calculateTotal()Errors:
    "c-kw">def test_raises_value_error_on_invalid_input(self):
        "c-kw">with "c-kw">pytest."c-kw">raises(ValueError):
            Function / Method Name: calculateTotal()('invalid')

    "c-kw">def test_raises_type_error_on_wrong_type(self):
        "c-kw">with "c-kw">pytest."c-kw">raises(TypeError):
            Function / Method Name: calculateTotal()(123)

    "c-kw">def test_error_message_is_descriptive(self):
        "c-kw">with "c-kw">pytest."c-kw">raises(ValueError, match=r'invalid|error'):
            Function / Method Name: calculateTotal()(None)
```

BOUNDARY VALUE TESTS

3 tests

```
"c-kw">class TestFunction / Method Name: calculateTotal()Boundaries:
    "c-kw">def test_minimum_boundary(self):
        "c-kw">assert Function / Method Name: calculateTotal()(0) is not None

    "c-kw">def test_maximum_boundary(self):
        "c-kw">assert Function / Method Name: calculateTotal()(2**31-1) is not None

    "c-kw">def test_just_below_minimum_raises(self):
        "c-kw">import "c-kw">pytest
        "c-kw">with "c-kw">pytest."c-kw">raises(ValueError):
            Function / Method Name: calculateTotal()(-1)
```

COVERAGE SUMMARY

86%

83%	75%	81%
LINES	BRANCHES	FUNCTIONS

⬡ AI ANALYSIS

WHAT THESE TESTS COVER

- Core functionality with valid inputs
- Edge cases including empty and large values
- Null and undefined parameter handling
- Exception throwing and error messages
- Boundary values at min and max ranges

WHAT IS NOT COVERED

- Mocked external dependencies
- Performance regression tests

RECOMMENDED ADDITIONS

→ Add integration tests for Function / Method Name: calculateTotal() with real database/service

→ Add snapshot tests if Function / Method Name: calculateTotal() returns complex objects

→ Consider property-based testing (e.g. fast-check) for Function / Method Name: calculateTotal()

Tester: **Sahana** Fri, 13 Mar 2026 07:53:00 GMT TEST-HQ4KO1 15 TESTS ~249ms